Report 2: UML Class Diagram Report

Project Title: School Management System

System Overview: This report details the UML Class Model for the School Management System, defining the classes, attributes, methods, and relationships between entities.

1. Classes, Attributes, and Methods

1. Student Class:

Attributes: studentId: int, fullName: String, email: String, dateOfBirth: Date

Methods: registerCourse(), viewGrades()

2. Teacher Class:

Attributes: teacherId: int, fullName: String, specialization: String, email: String

Methods: assignGrade(), createCourse()

3. Course Class:

Attributes: courseId: int, courseName: String, credits: int

Methods: getCourseDetails()

4. Enrollment Class (Junction Class):

Attributes: enrollmentId: int, enrollmentDate: Date, status: String

Methods: updateStatus()

2. Relationships & Multiplicities

Teacher to Course (One-to-Many 1 .. *):

One teacher can teach one or multiple courses.

A course is taught by exactly one teacher.

Student to Enrollment (One-to-Many 1 .. *):

One student can have multiple enrollments (register for multiple courses).

Each enrollment record belongs to exactly one student.

Course to Enrollment (One-to-Many 1 .. *):

One course can have multiple student enrollments.

Each enrollment record belongs to a specific course.

(Note: The combination of Student, Course, and Enrollment represents the Many-to-Many relationship between Students and Courses).

3. UML Class Diagram

